

Curriculum Vitae



Gyumin Lee

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POSITION HELD

- **Dongguk University** Seoul, Republic of Korea
Department of Management Information Systems Sep. 2024 – Present
Assistant Professor
- **Korea University** Seoul, Republic of Korea
Department of Statistics Mar. 2024 – Aug. 2024
Postdoctoral Researcher
- **Korea University** Seoul, Republic of Korea
Center for AI and Big Data in Public Sector, Aug. 2023 – Aug. 2024
Institute of Governmental Studies
Participating researcher
- **Ulsan National Institute of Science and Technology** Ulsan, Republic of Korea
Graduate School of Interdisciplinary Management Aug. 2023 – Jan. 2024
Collegiate faculty

EDUCATION

- **Ulsan National Institute of Science and Technology** Ulsan, Republic of Korea
School of Business Administration Aug. 2017 – Aug. 2023
Doctor of Philosophy in Management Engineering
Thesis: "Development of decision support systems for technology and business opportunity analysis"
- **Ulsan National Institute of Science and Technology** Ulsan, Republic of Korea
School of Electrical and Computer Engineering Mar. 2011 – Aug. 2017
Bachelor of Engineering in Computer Science

RESEARCH EXPERIENCE

- Innovation and Policy Analytics LAB (IPA-LAB) *Aug. 2023 – Aug. 2024*
Researcher
- Technology and Innovation Management Lab (TIM-LAB) *Aug. 2017 – Aug. 2023*
Research Assistant
- Technology and Innovation Management Lab (TIM-LAB) *Jun. 2016 – Aug. 2017*
Undergraduate Research Assistant

RESEARCH INTERESTS

- Business analytics
- Technology management
- Intelligent manufacturing systems
- Drug discovery
- Natural language processing
- Generative AI

SKILLS AND TECHNIQUES

- Key skills: Machine learning, Deep learning, Language model, System design and development, Web scraping
- Technical skills: Python, R, SQL

PUBLICATIONS

Peer Reviewed Journal Articles

1. **Gyumin Lee** and Changyong Lee, 2025. Technological impact-guided technology opportunity analysis using a generative-predictive machine learning model. *Scientometrics (SCIE/SSCI)*, 131, p.2439–2469. DOI: 10.1007/s11192-025-05491-z. ISSN: 0138-9130.
[Published on Dec. 6, 2025]
2. Jungwon Park, **Gyumin Lee**, Daeseong Jeon, Jaewoong Choi, and Changyong Lee, 2024. Language models and administrative innovation in the public sector: Concept, approaches, and considerations. *Journal of Governmental Studies (KCI)*. DOI: 10.19067/jgs.2024.30.1.35. ISSN: 1229-4241.
[Published on Apr. 17, 2024]
3. **Gyumin Lee**, Sungjun Lee, and Changyong Lee, 2023. Inventor–licensee matchmaking for university technology licensing: A fastText approach. *Technovation (SCIE/SSCI)*, 125, p.102765. DOI: 10.1016/j.technovation.2023.102765. ISSN: 0166-4972.
[Published on Jul. 1, 2023]
4. **Gyumin Lee**, Daeil Kwon, and Changyong Lee, 2023. A convolutional neural network model for SOH estimation of Li-ion batteries with physical interpretability. *Mechanical Systems and Signal Processing (SCIE)*, 188, p.110004. DOI: 10.1016/j.ymssp.2022.110004. ISSN: 0888-3270.
[Published on Apr. 1, 2023]
5. Joram Kim, **Gyumin Lee**, Seungbin Lee, and Changyong Lee, 2022. Towards expert–machine collaborations for technology valuation: An interpretable machine learning approach. *Technological Forecasting and Social Change (SSCI)*, 183, p.121940. DOI: 10.1016/j.techfore.2022.121940. ISSN: 0040-1625.
[Published on Oct. 1, 2022]
6. **Gyumin Lee**, Joram Kim, and Changyong Lee, 2022. State-of-health estimation of

- Li-ion batteries in the early phases of qualification tests: An interpretable machine learning approach. *Expert Systems with Applications (SCIE)*, 197, p.116817. DOI: 10.1016/j.eswa.2022.116817. ISSN: 0957-4174. [Published on Jul. 1, 2022]
7. **Gyumin Lee**, Seung Jun Lee, and Changyong Lee, 2021. A convolutional neural network model for abnormality diagnosis in a nuclear power plant. *Applied Soft Computing (SCIE)*, 99, p.106874. DOI: 10.1016/j.asoc.2020.106874. ISSN: 1568-4946. [Published on Feb. 1, 2021]
 8. Jae Min Kim, **Gyumin Lee**, Changyong Lee, and Seung Jun Lee, 2020. Abnormality diagnosis model for nuclear power plants using two-stage gated recurrent units. *Nuclear Engineering and Technology (SCIE)*, 52(9), p.2009–2016. DOI: 10.1016/j.net.2020.02.002. ISSN: 1738-5733. [Published on Sep. 1, 2020]
 9. **Gyumin Lee**, Daejin Kim, and Changyong Lee, 2020. A sequential pattern mining approach to identifying potential areas for business diversification. *Asian Journal of Technology Innovation (SSCI)*, 28(1), p.21–41. DOI: 10.1080/19761597.2019.1693900. ISSN: 1976-1597. [Published on Jan. 2, 2020]
 10. Changyong Lee and **Gyumin Lee**, 2019. Technology opportunity analysis based on recombinant search: patent landscape analysis for idea generation. *Scientometrics (SCIE/SSCI)*, 121(2), p.603–632. DOI: 10.1007/s11192-019-03224-7. ISSN: 0138-9130. [Published on Nov. 1, 2019]

PROJECTS

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| 1. Development of a method for measuring similarity between designated products
Korea Institute of Intellectual Property (KIIP, 한국지식재산연구원) | May. 2023 – Nov. 2023 |
| 2. Planning of a system for diagnosing industrial innovation capability
Korea Planning & Evaluation Institute of Industrial Technology (KEIT, 한국산업기술기획평가원) | Jan. 2023 – Aug. 2023 |
| 3. Development and application of methods towards the NRF's digital transformation: Focusing on 'expert panel organization' and 'researcher activity analysis'
National Research Foundation of Korea (NRF, 한국연구재단) | Nov. 2022 – May. 2023 |
| 4. Railway power distribution line sensing data analysis
Nemosis ((주)네모시스) | Mar. 2022 – Oct. 2022 |
| 5. Developing a concordance table for integrated IP analytics: Linkages among patent, trademark, and design classification codes
Korea Intellectual Property Office (KIPO, 특허청) | Jun. 2021 – Jan. 2022 |
| 6. Development of a model for shopping products classification and consultation intention classification
KTH | Dec. 2020 – Mar. 2021 |
| 7. Development of a battery life deterioration prediction method to accelerate qualification tests on Li-ion batteries
Hyundai Motor Group (현대자동차) | Mar. 2020 – Sep. 2020 |
| 8. Discovery of Serendipitous drug candidates based on artificial | Jan. 2019 – Aug. 2023 |

intelligence methods

Daewoong Foundation (대웅재단)

9. Development of operation supporting technology based on artificial intelligence for nuclear power plant start up and shutdown operation *Jan. 2018 – Mar. 2019*

Korea Hydro & Nuclear Power (KHNP, 한국수력원자력)

10. Identifying high-risk subrogation group based on the analysis of life stability funds *Oct. 2016 – Dec. 2016*

Korea Workers' Compensation and Welfare Service (KCOMWEL, 근로복지공단)

CONFERENCES

1. Changhun Lee and **Gyumin Lee**, 2024. Repurformer: Transformers for repurposing-aware molecule generation. In *Language and Molecules ACL 2024 (Oral)*.
2. **Gyumin Lee** and Changyong Lee, 2024. Seek your fortune!: Technological impact-guided technology opportunity analysis using generative-predictive machine learning models. In *2024 Winter Conference of Korea Society for Innovation Management & Economics*.
3. **Gyumin Lee** and Changhun Lee, 2021. Development of a method for identifying new drug candidates based on generative and predictive deep learning models considering the possibility of drug repurposing. In *2021 Spring Conference of Korean Institute of Industrial Engineering*.
4. Jae Min Kim, **Gyumin Lee**, Suckwon Hong, and Seung Jun Lee, 2019. Input data dimensionality reduction of abnormality diagnosis model for nuclear power plants. In *International Conference on Intelligent Human Systems Integration*.
5. Changhun Lee, **Gyumin Lee**, and Chiehyeon Lim, 2018. Development of a context-aware serendipitous recommender system. In *INFORMS Annual Meeting 2018*.

COURSES TAUGHT

- | | |
|---|------------|
| 1. [UG] AI Trends and Business Innovation | 2025 |
| 2. [UG] Generative AI and Business Innovation | 2025, 2026 |
| 3. [UG] Deep Learning | 2024, 2025 |
| 4. [UG] Applied AI | 2025, 2026 |
| 5. [UG] Principle of Management Information Systems | 2024 |
| 6. [Graduate] Management Information Systems Seminar | 2026 |
| 7. [Graduate] Machine Learning and Deep Learning | 2025 |
| 8. [Graduate] Advanced Data Mining with AI | 2023 |
| 9. [Graduate] Database Management | 2023 |
| 10. [MBA] Principle of Management Information Systems | 2025 |
| 11. [MBA] Programming for Business Data Analytics | 2025, 2026 |

HONORS AND AWARDS

1. **UNIST Best Research Award**, *Feb. 2024*
Ulsan National Institute of Science and Technology
In recognition of outstanding performance in the conduct of research.
Thesis titled "Development of decision support systems for technology and business opportunity analysis"